Greater Sydney SA2 Well-Resourced Analysis

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1. Introduction

Objective: To develop a metric for assessing the "well-resourced" of each SA2 region in Greater Sydney by integrating various publicly accessible datasets that provide information on geographical characteristics.

Selected SA4:

- Sydney - North Sydney and Hornsby
- Sydney - City and Inner South
- Sydney - Inner South West

Selected industries:

- Mining
- Construction
- Manufacturing

2. Dataset Description

Summarize each dataset:
See Table 1 in appendix

Prior to analysis, all datasets underwent rigorous validation to assess data completeness. Missing values were systematically identified using Python's isnull() and sum() functions.

After validation, it was determined that missing values:

- Were confined to non-essential columns
- Had a minimised impact on aggregate metrics
- Did not affect score calculation
- POI dataset formation

Thus, no imputation was required, and incomplete entries were safely excluded to preserve analytical rigor.

2.1 Preprocessing Steps

SA2 Boundaries: We imported Statistical Area 2 (SA2) boundaries for Australia, then filtered to retain only regions within Greater Sydney. The spatial data was converted to the GDA94 coordinate system (EPSG:4283) and simplified to retain key attributes. We verified all polygon geometries were valid and confirmed the coordinate system.

Catchments: Three school catchment datasets (future, primary, and secondary) were combined into a single dataset. We standardised the coordinate system to match our other spatial data (EPSG:4283) and retained only key attributes. A zero buffer operation was applied to resolve any geometric inconsistencies, followed by validation checks for geometry integrity.

Stops: Public transport stop locations (geometry) were created from latitude and longitude coordinates in the CSV file. These points were converted to spatial features using the same coordinate system as our other datasets (EPSG:4283). We maintained stop id, geometry and names and spatial validity is also verified.

Businesses: Business data was filtered to focus on three key industries: Mining, Manufacturing, and Construction. We aggregated business counts to SA2 regions, ensuring each record shows total businesses per industry per geographic area.

Population: Young people column is created through summation of all age groups. We excluded SA2 regions with fewer than 100 residents.

Income: Income records were simplified to retain median income and corresponding SA2 identifiers. This created a straightforward dataset for linking economic characteristics to geographic regions in subsequent analysis.

POI: We extracted Points of Interest by iterating over each SA2 region's bounding box. For each SA2 we constructed an Esri 'envelope' geometry, called the NSW POI ArcGIS REST API (with optional API key) at that box, paused 1 second between calls, and collected GeoJSON features into a GeoDataFrame before persisting to PostGIS.

3. Database Description

3.1 Schema Design

See figure 1 in appendix

3.2 Table Linkages

sa2_regions is the parent with a primary key, SA2_CODE21. Businesses, population and income each carry a foreign key column, either sa2_code or sa2_code21, that references sa2_regions's primary key. Stops and schools are not linked through it, but joined to sa2_regions spatially.

3.3 Indexing Strategy

Two major types of index are used, GIST spatial index and B-Tree index. Tables with geometry values such as sa2_regions, stops, and schools have GIST spatial indexes created for them. sa2_geom_idx is created for sa2_regions, stops_geom_idx is created for stops, and schools_geom_idx is created for schools. Since businesses table does not contain a geometry column, a B-Tree index is used for it. An index, businesses_sa2_idx, is created for it to join keys. No indexes are created for income and population because they have foreign keys linking to sa2_regions, and PostgreSQL automatically created unique B-Tree indexes on them.

4. Score Analysis

Scoring Formula

$$\text{Score} = S(z_{\text{business}} + z_{\text{stops}} + z_{\text{schools}} + z_{\text{POI}})$$

Where S is the sigmoid function and z are z-scores.
Sigmoid function S converts the summed z-scores into a value between 0 and 1 for interpretability:

$$S(x) = \frac{1}{1 + e^{-x}}$$

Normalization: Z-scores ensure metrics are comparable. For example, businesses and schools can be compared.
Outlier Mitigation: The sigmoid function bounds scores between 0 and 1, which compresses extreme values.
Customization: Industries and POI groups are selected based on relevance.
Filtering: SA2 regions with populations less than 100 are excluded to avoid skewed per-capita calculations.

Each component plays a crucial role in the formula. Businesses reflect economic activity, stops measure public transport accessibility, schools indicate educational resources for young populations, and POIs represent amenities. They are all significant to determine if a place is “well-resourced”.

Results (see figure 4&5 in appendix)

Top SA2s: Millers Point (Score ≈ 1.0)
Bottom SA2s: Port botany Industrial (Score ≈ 0.02)

Millers Point’s high score is driven by its exceptional business density (z = 7.08) and POI availability (z = 2.87), likely reflecting its mix of commercial, residential, and historical landmarks in a prime urban location. The area is not purely industrial which means it features many offices, heritage sites, and waterfront amenities, creating a high concentration of businesses and POIs such as restaurants and cultural attractions. However, its lower z-scores for schools (-0.49) and modest public transport (0.64) suggest limited educational infrastructure and average transit connectivity. The score prioritizes economic activity and amenities, masking weaknesses in services like schools, which are less critical in a densely developed urban zone dominated by working professionals and tourism.

Port Botany Industrial’s low score reflects its industrial specialization, which prioritizes logistical infrastructure over residential amenities. Its below-average z-scores for public transport stops (-1.37), schools per young people (-1.42), and POIs (-1.22) indicate minimal services for residents, typical of industrial zones where factories, warehouses, and freight hubs dominate. While businesses per 1000 people (z=0) align with the regional average, these are likely industrial enterprises rather than community-facing services. The area's focus on heavy industry limits access to schools, public transit, and recreational/cultural POIs, resulting in a low "well-resourced" score despite its economic role.

SA4 Comparison:

- North Sydney and Hornsby (Average Score ≈ 0.56)
- City and Inner South (Average Score ≈ 0.27)
- Inner South West (Average Score ≈ 0.50)

North Sydney and Hornsby (Average Score ≈ 0.56)
This region's high score reflects its well-serviced suburbs, where good commercial strips and mixed land use create pedestrian-friendly neighbourhoods with convenient access to amenities.

Inner South West (Average Score ≈ 0.50)
The well-balanced average score of the region comes from the uniform combination of residential, commercial, and parks like those found in suburbs. Main transport infrastructure like the M4 Motorway and Burwood Station contributes to convenience, but deteriorating infrastructure in a few pockets lowers the efficiency of public transport. Suburban gentrification compels the shift from industrial to mixed-use spaces, generating uneven amenity distribution—fresh cafes and cultural hubs alongside slender schools and services.

City and Inner South (Average Score ≈ 0.27
Low score for this area is attributed to extreme spatial polarization. Logistic and manufacturing precincts like Port Botany are mainly for manufacturing and logistics but not for residential amenities. Conversely, highly populated city centers like Millers Point score well owing to tourism and commercial density but possess crowded transport hubs and limited parks. The formula's preference for businesses and POIs penalizes solely industrialized or hyper-urban areas, emphasizing the need for strategies that better represent equity and community services.

Interesting findings:
The City and Inner South region includes some of Sydney's most iconic areas, such as Pyrmont, Haymarket, and Millers Point—places often considered the heart of the city. Generally, people may think it must have a very high score. Surprisingly, it actually has the lowest average score among the three SA4 regions. It is because these places are not mainly for industries like manufacturing, and construction. That's why a region home to landmarks like Darling Harbour, the Sydney Tower Eye, and the Opera House would rank the lowest overall.

Visualizations

See Figure 2 in appendix

Scrutinising the results

Based on the formula of calculating the score, we can conclude that the higher the total z-score, the higher the score. Thus, total z-score is directly proportional to score. Based on our result (see figure 6 in appendix), we notice that the z-score of the number of POI is the highest on average with a value of 1.415851e-18. Thus, it is concluded that the POI component played a significant role in the selected area.

5. Correlation with Income

Result

Based on the results of pearson and spearman correlation, we have:

- Score: Pearson = -0.094334
    - The score shows weak and insignificant correlations which indicates that the composite “well-resourced” metric does not strongly align with the median income. Since the selected industries, manufacturing and construction are all heavy industries which are usually located in rural areas. In rural areas, median incomes tend to be lower. This could be due to factors like seasonal employment, lower educational attainment, or fewer high-paying urban service-sector jobs. While retail trade businesses are everywhere. However, median income alone may not have enough impact in these regions. For example, people in heavy industries might have high wages but small populations, skewing the median. However, workers in retail trade tend to have lower median income. The weak correlation implies that median income is not a dominant driver of the score. Future iterations of the scoring formula could include additional socioeconomic variables like healthcare access to better reflect "well-resourced" status in rural contexts.

Scatter Plot

See figure 3 in appendix

6. Limitations

The score’s limitations arise from its overemphasis on infrastructure and amenity density such as public transport, schools, businesses, and POIs, while neglecting socioeconomic factors like healthcare access, which disproportionately skew results toward urban areas. The score is biased toward cities and doesn’t reflect rural areas well. In the rural areas, small populations and big industries can make data look better per person, even if incomes are lower. Giving equal weight to all factors assumes they matter the same everywhere, but some things like public transport are less important in rural areas.

Appendix

SQL Snippets:

```
DROP TABLE IF EXISTS income CASCADE;
DROP TABLE IF EXISTS population CASCADE;
DROP TABLE IF EXISTS schools CASCADE;
DROP TABLE IF EXISTS stops CASCADE;
DROP TABLE IF EXISTS businesses CASCADE;
DROP TABLE IF EXISTS sa2_regions CASCADE;

CREATE TABLE sa2_regions (
   SA2_CODE21    VARCHAR(50)  PRIMARY KEY,
   SA2_NAME21    VARCHAR(255),
   SA4_CODE21    VARCHAR(50),
   SA4_NAME21    VARCHAR(255),
   geometry      GEOMETRY(Geometry, 4283)
);
CREATE INDEX sa2_geom_idx ON sa2_regions USING GIST (geometry);

CREATE TABLE businesses (
   sa2_code        VARCHAR(50)  REFERENCES sa2_regions(SA2_CODE21),
   sa2_name        VARCHAR(255),
   industry_name   VARCHAR(255),
   total_businesses INTEGER
);
CREATE INDEX businesses_sa2_idx ON businesses (sa2_code);

CREATE TABLE stops (
   stop_id   VARCHAR(50)    PRIMARY KEY,
   stop_name VARCHAR(255),
   geometry  GEOMETRY(Point, 4283)
);
CREATE INDEX stops_geom_idx ON stops USING GIST (geometry);

CREATE TABLE schools (
   USE_ID    VARCHAR(50)      PRIMARY KEY,
   geometry  GEOMETRY(MultiPolygon, 4283)
);
CREATE INDEX schools_geom_idx ON schools USING GIST (geometry);

CREATE TABLE population (
```

```
    sa2_code        VARCHAR(50)    PRIMARY KEY REFERENCES
sa2_regions(SA2_CODE21),
  sa2_name    VARCHAR(255),
  total_people  INTEGER,
  young_people  INTEGER
);

CREATE TABLE income (
    sa2_code21      VARCHAR(50)    PRIMARY KEY REFERENCES
sa2_regions(SA2_CODE21),
  sa2_name21   VARCHAR(255),
  median_income NUMERIC
);
```

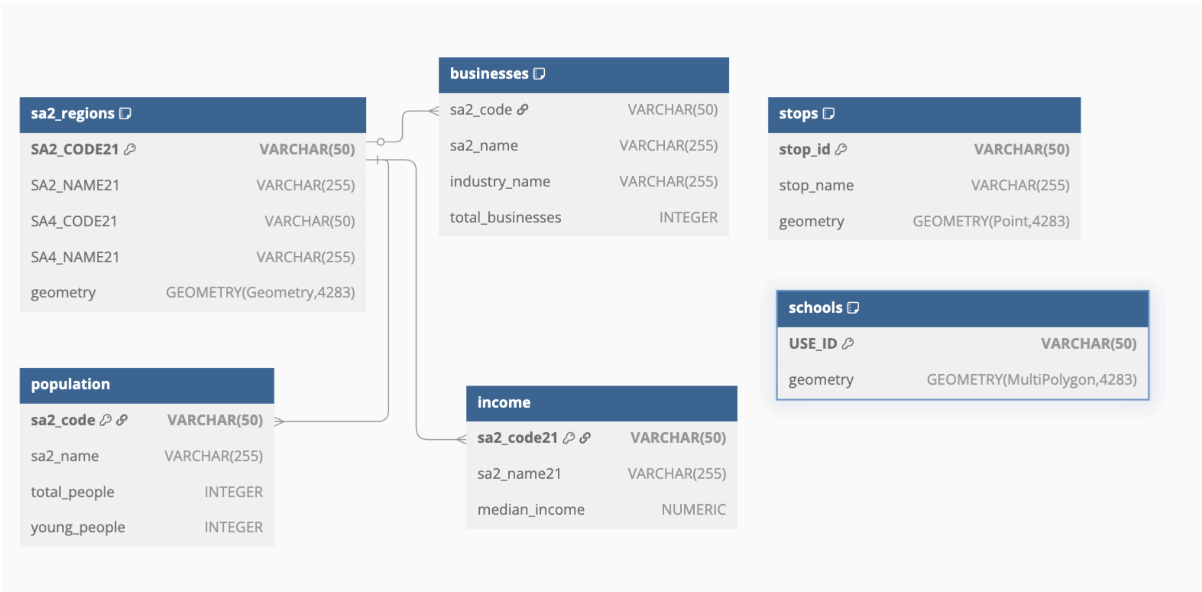
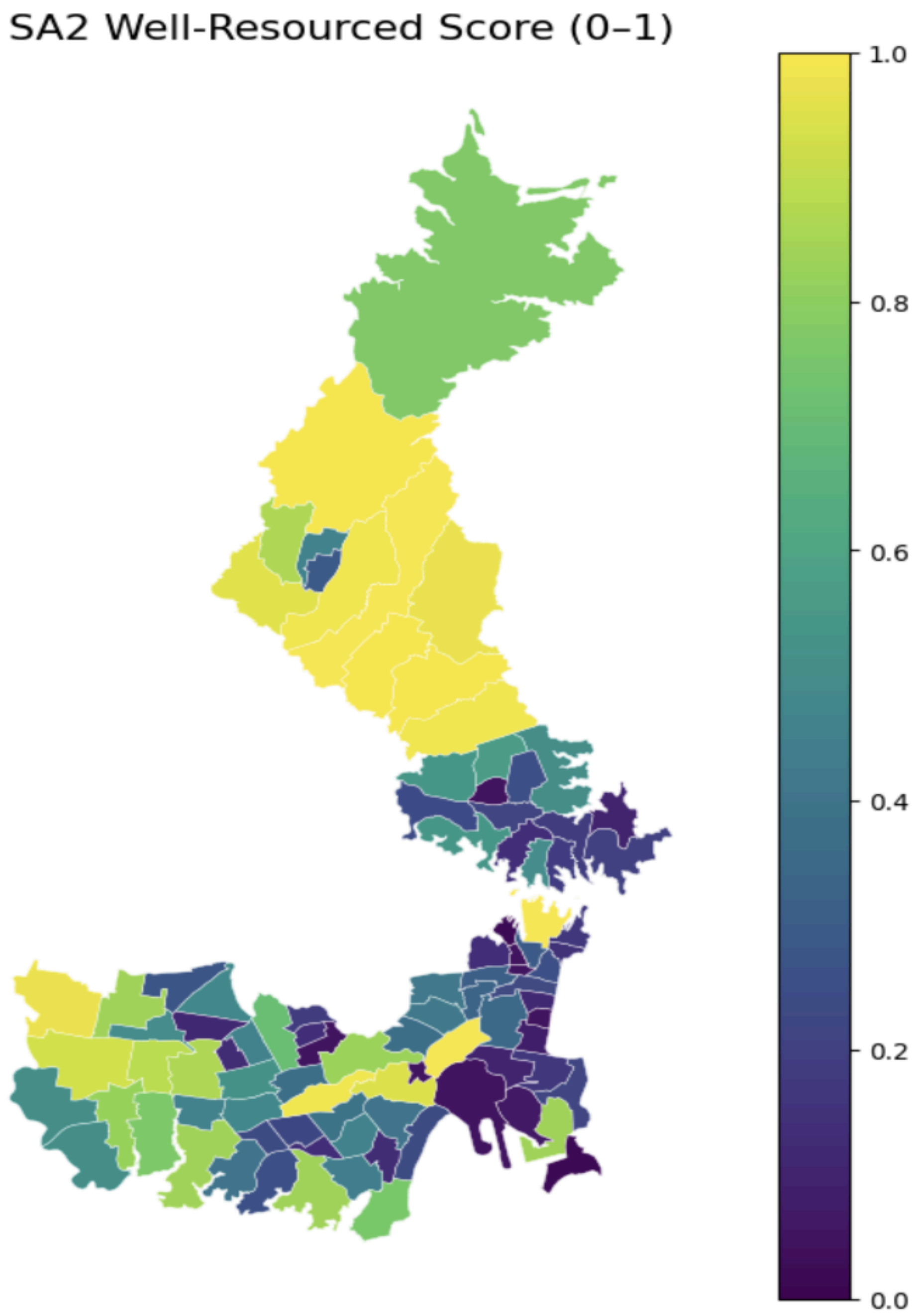


Figure 1

Dataset	Source	Key Attributes
SA2 Boundaries	[ABS]	[SA2_CODE21] [SA2_NAME21] [SA4_CODE21] [SA4_NAME21]
Catchments	[NSW Department of Education]	[USE_ID] [CATCH_TYPE] [geometry]
Businesses	[Australian Bureau of Statistics]	[sa2_code] [sa2_name]

		[industry_name] [total_businesses]
Stops	Transport for NSW	[stop_id] [stop_name] [geometry]
Income	Canvas	[sa2_code21] [sa2_name] [median_income]
Population	Canvas	[sa2_code] [sa2_name] [total_people] [young_people]
POI	NSW Points of Interest API	[OBJECTID] [sa2_code] [SA2_CODE21]

Table 1



The score of Each Components of Millers Point and Port Botany Industrial

Figure 2

Figure 3

Highhest and lowest score

```
df_scores = pd.read_sql('SELECT sa2_name, sa4_name, score FROM sa2_scores ORDER BY score DESC LIMIT 1', con=db)
print(df_scores)

df_scores = pd.read_sql('SELECT sa2_name, sa4_name, score FROM sa2_scores ORDER BY score ASC LIMIT 1', con=db)
print(df_scores)
```

```
                         sa2_name                        sa4_name     score
0  Sydney (North) - Millers Point  Sydney - City and Inner South  0.999959
                 sa2_name                        sa4_name     score
0  Port Botany Industrial  Sydney - City and Inner South  0.017775
```

Average Score of SA4

```
df_scores = pd.read_sql('SELECT sa4_name, AVG(score) FROM sa2_scores GROUP BY sa4_name', db)
print(df_scores)
```

```
                         sa4_name       avg
0          Sydney - Inner South West  0.500807
1      Sydney - City and Inner South  0.266692
2  Sydney - North Sydney and Hornsby  0.555771
```

Figure 4

The score of Each Components of Millers Point and Port Botany Industrial

```python
z_df = pd.read_sql(
    """
    SELECT
      z_business_per_1000 AS avg_z_business_per_1000,
      z_num_stops AS avg_z_num_stops,
      z_schools_per_1000 AS avg_z_schools_per_1000,
      z_num_poi AS avg_z_num_poi
    FROM sa2_scores
    where sa2_name = 'Sydney (North) - Millers Point' or sa2_name = 'Port Botany Industrial'
    """,
    con=db
)

display(z_df)
```

	avg_z_business_per_1000	avg_z_num_stops	avg_z_schools_per_1000	avg_z_num_poi
0	7.077685	0.639161	-0.492947	2.872394
1	0.000000	-1.367698	-1.424345	-1.219983

Figure 5

Average z-score

```python
avg_z_df = pd.read_sql(
    """
    SELECT
      AVG(z_business_per_1000) AS avg_z_business_per_1000,
      AVG(z_num_stops) AS avg_z_num_stops,
      AVG(z_schools_per_1000) AS avg_z_schools_per_1000,
      AVG(z_num_poi) AS avg_z_num_poi
    FROM sa2_scores
    """,
    con=db
)

avg_z_df.index = ["Average"]

print("Average z-scores:")
display(avg_z_df)
```

```
Average z-scores:
```

	avg_z_business_per_1000	avg_z_num_stops	avg_z_schools_per_1000	avg_z_num_poi
Average	-4.519087e-23	9.521277e-19	-1.886064e-18	1.415851e-18

Figure 6